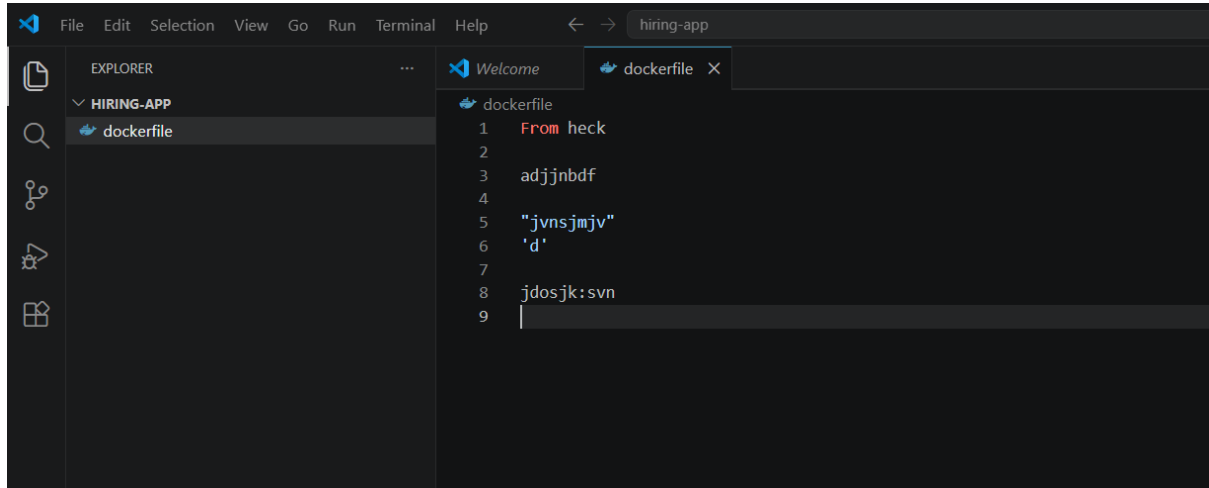


Git and GitHub -01

- Create a folder in desktop and open in vs code



- And create File name as dockerfile

Name	Date modified	Type	Size
dockerfile	07-04-2026 15:19	File	1 KB

- open in githbash and install git version

```
Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app
$ git --version
git version 2.37.1.windows.1

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app
$
```

- Check the list file (ls) where dockerfile is available or not

```
Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app
$ ls
dockerfile

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app
$ git status
fatal: not a git repository (or any of the parent directories): .git

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app
$
```

- Initializing the git

```
Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app
$ ls -a
./ ../ dockerfile

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app
$ git init
Initialized empty Git repository in C:/Users/LENOVO/Desktop/hiring-app/.git/

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$
```

- .git/

```
Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ cd .git/

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app/.git (GIT_DIR!)
$ ll
total 7
-rw-r--r-- 1 Arunsai 197121 23 Apr 7 15:24 HEAD
-rw-r--r-- 1 Arunsai 197121 130 Apr 7 15:24 config
-rw-r--r-- 1 Arunsai 197121 73 Apr 7 15:24 description
-rwxr-xr-x 1 Arunsai 197121 0 Apr 7 15:24 hooks/
-rwxr-xr-x 1 Arunsai 197121 0 Apr 7 15:24 info/
-rwxr-xr-x 1 Arunsai 197121 0 Apr 7 15:24 objects/
-rwxr-xr-x 1 Arunsai 197121 0 Apr 7 15:24 refs/

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app/.git (GIT_DIR!)
$
```

- Git status of dockerfile

```
Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ ls -a
./ ../ .git/ dockerfile

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    dockerfile

nothing added to commit but untracked files present (use "git add" to track)

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$
```

- Remove .git/

```
Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ ls -a
./ ../ .git/ dockerfile

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ rm -rf .git/

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app
$ git status
fatal: not a git repository (or any of the parent directories): .git

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app
$ git init
Initialized empty Git repository in C:/Users/LENOVO/Desktop/hiring-app/.git/

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$
```

- Git config – list

```
Arunasai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ git config --list
diff.astextplain.textconv=astextplain
filter.lfs.clean=git-lfs clean -- %f
filter.lfs.smudge=git-lfs smudge -- %f
filter.lfs.process=git-lfs filter-process
filter.lfs.required=true
http.sslbackend=openssl
http.sslcainfo=C:/Program Files/Git/mingw64/ssl/certs/ca-bundle.crt
core.autocrlf=true
core.fscache=true
core.symlinks=false
pull.rebase=false
credential.helper=manager-core
credential.https://dev.azure.com.usehttppath=true
init.defaultbranch=master
core.repositoryformatversion=0
core.filemode=false
core.bare=false
core.logallrefupdates=true
core.symlinks=false
core.ignorecase=true
```

- Create User name and email in git

```
Arunasai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ git config --global user.name "Arunasai"

Arunasai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ git config --list
diff.astextplain.textconv=astextplain
filter.lfs.clean=git-lfs clean -- %f
filter.lfs.smudge=git-lfs smudge -- %f
filter.lfs.process=git-lfs filter-process
filter.lfs.required=true
http.sslbackend=openssl
http.sslcainfo=C:/Program Files/Git/mingw64/ssl/certs/ca-bundle.crt
core.autocrlf=true
core.fscache=true
core.symlinks=false
pull.rebase=false
credential.helper=manager-core
credential.https://dev.azure.com.usehttppath=true
init.defaultbranch=master
user.email=arunsaimailme17@gmail.com
user.name=Arunasai
core.repositoryformatversion=0
core.filemode=false
core.bare=false
core.logallrefupdates=true
core.symlinks=false
core.ignorecase=true
```

```
Arunasai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ git config --global user.email "arunsaimailme17@gmail.com"

Arunasai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ |
```

- Git status

```
Arunasai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   dockerfile

Arunasai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$
```

- Creating new data in git or files

```
Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ ls
b c d dockerfile e f g h i j

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   dockerfile

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    b
    c
    d
    e
    f
    g
    h
    i
    j
```

- By using add . we can create all files at a time

```
Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ git add .

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   b
    new file:   c
    new file:   d
    new file:   dockerfile
    new file:   e
    new file:   f
    new file:   g
    new file:   h
    new file:   i
    new file:   j

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$
```

- By using git commit -m “adding couple of files”

```
Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ git commit -m "added couple of files"
[master (root-commit) 7100e83] added couple of files
10 files changed, 8 insertions(+)
create mode 100644 b
create mode 100644 c
create mode 100644 d
create mode 100644 dockerfile
create mode 100644 e
create mode 100644 f
create mode 100644 g
create mode 100644 h
create mode 100644 i
create mode 100644 j

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ .....
```

- To check git log

```
Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ git log
commit 83e6c185f3fea21b55a6a5de9464d22fe9ece7ed (HEAD -> master)
Author: Arunsai <arunsaimailme17@gmail.com>
Date: Tue Apr 7 15:58:09 2026 +0530

    added couple of file

commit 7100e8342b45e46be420725e142ee9e4652522
```

- Add new data by using touch x

```
Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ touch x

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ git add x

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ git commit -m "added x to the project"
[master 1e83d74] added x to the project
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 x

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ git log
commit 1e83d74c82106a6280548e4325ea6e0ac72066e3 (HEAD -> master)
Author: Arunsai <arunsaimailme17@gmail.com>
Date: Tue Apr 7 16:11:55 2026 +0530

    added x to the project

commit 025dc9e6619bff8a2c9bf76c47cb1784b3a9b39d
Author: Arunsai <arunsaimailme17@gmail.com>
Date: Tue Apr 7 15:39:05 2026 +0530

    added couple of files
```

- Create a github and created a repository as techie-horizon 05



Create a new repository

Repositories contain a project's files and version history. Have a project elsewhere? [Import a repository](#).
Required fields are marked with an asterisk (*).

1 General

Owner * Arunsai873 / Repository name * techie-horizon

techie-horizon is available.

- And added the github SSH link in gitbash for git remote

```
Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ git remote -v

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ git remote add origin git@github.com:Arunsai873/techie-horizon-05.git

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ git remote -v
origin  git@github.com:Arunsai873/techie-horizon-05.git (fetch)
origin  git@github.com:Arunsai873/techie-horizon-05.git (push)

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$
```

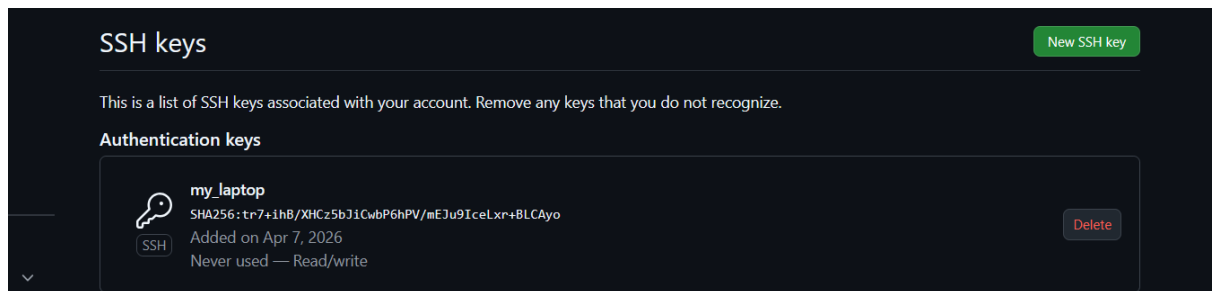
- Get the SSH-Keyegen

```
Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ ssh-keygen
Generating public/private rsa key pair.
Enter file in which to save the key (/c/Users/LENOVO/.ssh/id_rsa):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /c/Users/LENOVO/.ssh/id_rsa
Your public key has been saved in /c/Users/LENOVO/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:tr7+ihB/XHCz5bJiCwbP6hPV/mEJu9IceLxr+BLCAyo Arunsai@DESKTOP-N6MPCBS
The key's randomart image is:
----[RSA 3072]-----+
      .. o .
     . . oo =
    . oo. +So+..
   E  =*oo*o+o
    . .+****.
     .++**o.
    .o..*O=.
----[SHA256]-----+

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$
```

- Get the SSH-keygen key and paste in github SSH keys and save it

```
Arunasai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ cat /c/Users/LENOVO/.ssh/id_rsa.pub
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQDeGcsEqxfZaLoSmK1Wv+LJFdFNGX8HkMO2sOnTA00h2ym1tcQt3dY8nKG
7rZNIu2dbcDhq6mhtgVrTPJa9eVuaUU47B1bMWDfd4ct4HPLsSJCawebh52svAKBY3zQcV/Ry0ahVdRfq+HEhSbARmHfFq9
Fy4wU4Rb7ehFDYnB3NySgrsCE+qJyod7LPVqvo06S5Zwnvj7EL2zfVnQavrNMH7qDxIjjdVrH6d5e6hrp10dI8xCVA2VV11
TRKsuwQJHRQL4EYrAwAkThLJqXLGxs6ZqgerL3Kn74AvtqALzDIv1w77QZ7yTFdvbMMqfyDUy1CDBr11pS7QswRLBLTbuM+
TzhZugjgo1hZczBto7b9nsywT41czDkE9pGxCxc51vaMfFpaQCrN1XfCBChBrj91jBbzTMP1YVcIzIMnVnfPwbiwtEe1kTt
L5Kc9rbd/MkjpnUhsJTMsmKJywyBvB9XLhLrZ05FQq+V3EkfYIH5Yaxh1yXSYSaj5/wWhyXupQU= Arunsai@DESKTOP-N
6MPCBS
```



- git push -u origin master

```
Arunasai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ git push -u origin master
The authenticity of host 'github.com (20.207.73.82)' can't be established.
ED25519 key fingerprint is SHA256:+DiY3wvV6TuJJhbpZisF/zLDA0zPMSvHdkr4UvCOQu.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'github.com' (ED25519) to the list of known hosts.
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 8 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (6/6), 533 bytes | 533.00 KiB/s, done.
Total 6 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), done.
To github.com:Arunasai873/techie-horizon-05.git
 * [new branch]      master -> master
branch 'master' set up to track 'origin/master'.

Arunasai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$
```

- remove dockerfile

```
Arunasai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ ls
a b c d dockerfile e f g h i j k m n p x z

Arunasai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ rm dockerfile

Arunasai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ ls
a b c d e f g h i j k m n p x z

Arunasai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$
```

- check in git status deleted dockerfile

```
Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ git status
On branch master
Your branch is up to date with 'origin/master'.

Changes not staged for commit:
  (use "git add/rm <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        deleted:    dockerfile

no changes added to commit (use "git add" and/or "git commit -a")

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$
```

- again, added dockerfile

```
Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ git add .

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ git commit -m "Deleted dockerfile"
[master 4818dfc] Deleted dockerfile
 1 file changed, 8 deletions(-)
 delete mode 100644 dockerfile

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ git push -u origin master
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (2/2), 226 bytes | 75.00 KiB/s, done.
Total 2 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:Arunsai873/techie-horizon-05.git
   1e83d74..4818dfc master -> master
branch 'master' set up to track 'origin/master'.
```

- created new file in github as techie

techie	file on central repo	now
--------	----------------------	-----

- created new touch abc

```
Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ git commit -m "added as file abc"
[master ba56011] added as file abc
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 abc

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ ls
a abc b c d e f g h i j k m n p x z

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ git push -u origin master
To github.com:Arunsai873/techie-horizon-05.git
 ! [rejected]        master -> master (fetch first)
error: failed to push some refs to 'github.com:Arunsai873/techie-horizon-05.git'
hint: Updates were rejected because the remote contains work that you do
hint: not have locally. This is usually caused by another repository pushing
hint: to the same ref. You may want to first integrate the remote changes
hint: (e.g., 'git pull ...') before pushing again.
hint: See the 'Note about fast-forwards' in 'git push --help' for details.

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$
```

- list to check abc files

```
Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ ls
a abc b c d e f g h i j k m n p techie x z

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
$ git push -u origin master
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 8 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (4/4), 478 bytes | 478.00 KiB/s, done.
Total 4 (delta 2), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (2/2), completed with 1 local object.
To github.com:Arunsai873/techie-horizon-05.git
   38d8dbd..43e025b  master -> master
branch 'master' set up to track 'origin/master'.

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/hiring-app (master)
```

- after deleting hiring -app in desktop and recycle bin get restore

```
Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop
$ git clone git@github.com:Arunsai873/techie-horizon-05.git
Cloning into 'techie-horizon-05'...
remote: Enumerating objects: 15, done.
remote: Counting objects: 100% (15/15), done.
remote: Compressing objects: 100% (8/8), done.
remote: Total 15 (delta 5), reused 12 (delta 4), pack-reused 0 (from 0)
Receiving objects: 100% (15/15), done.
Resolving deltas: 100% (5/5), done.

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop
$ ls
Angular/                               'New Text Document (3).TXT'
Arun/                                  'New Text Document (4).txt'
'Arun programs'/                       'New Text Document.TXT'
ArunResumeEx.pdf                      Portfolio/
'CLG Photos'/                          Practice/
'Counter Application'/                 Python/
DSA/                                    SBMS/
Devops/                                Spring/
EPFO/                                  TajMahal/
Firefox.exe*                           Teams.exe*
'Free Movies & Series websites.txt'    Thoughts/
Hibernate/                             Training.TXT
HitFilm/                               'Visual Studio Code.lnk'*
Images/                                'Weather App'/
Img/                                    winSCP.lnk*
JRTP/                                  Yugas/
Jyothi/                                a9/
Kane.lacddb                            bsx-accounts.csv
Kathi/                                 desktop.ini
Movies/                                'old hdd'/
'My Folder'/                           photos/
'Mytics Wars'/                         shoot/
Netflix/                               techie-horizon-05/
'New Text Document (2).TXT'

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop
$ cd techie-horizon-05

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/techie-horizon-05 (master)
$ ls
a abc b c d e f g h i j k m n p techie x z
```

- To check last deleted dockerfile

```

commit 43e025b3a47ed52143791b709c0b017bd71797db (HEAD -> master, origin/master, origin/HEAD)
merge: ba56011 38d8dbd
author: Arunsai <arunsaimailme17@gmail.com>
date: Tue Apr 7 17:09:40 2026 +0530

    Merge branch 'master' of github.com:Arunsai873/techie-horizon-05

commit ba56011d8ecb534dd1cfd8bc698cb3576d736e1f
author: Arunsai <arunsaimailme17@gmail.com>
date: Tue Apr 7 16:57:49 2026 +0530

    added as file abc

commit 38d8dbd756ef8529a030815fa577a1a41240ebfc
author: Arunsai <arunsaimailme17@gmail.com>
date: Tue Apr 7 16:56:12 2026 +0530

    file on central repo

commit 4818dfc84309e993507fb68157f9fd93e2e33cf9
author: Arunsai <arunsaimailme17@gmail.com>
date: Tue Apr 7 16:46:12 2026 +0530

    Deleted dockerfile

```

- Git revert

```

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/techie-horizon-05 (master)
$ git revert 4818d
[master cd7c1d0] Revert "Deleted dockerfile"
1 file changed, 8 insertions(+)
create mode 100644 dockerfile

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/techie-horizon-05 (master)
$ ls
a abc b c d dockerfile e f g h i j k m n p techie x z

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/techie-horizon-05 (master)
$

```

- Creating a alphabet file and check status and reset

```
Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/techie-horizon-05 (master)
$ git status
On branch master
Your branch is ahead of 'origin/master' by 1 commit.
  (use "git push" to publish your local commits)

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        alphabet

nothing added to commit but untracked files present (use "git add" to track)

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/techie-horizon-05 (master)
$ git add .

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/techie-horizon-05 (master)
$ git status
On branch master
Your branch is ahead of 'origin/master' by 1 commit.
  (use "git push" to publish your local commits)

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   alphabet
```

- Check with reset

```
Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/techie-horizon-05 (master)
$ git reset

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/techie-horizon-05 (master)
$ git status
On branch master
Your branch is ahead of 'origin/master' by 1 commit.
  (use "git push" to publish your local commits)

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        alphabet

nothing added to commit but untracked files present (use "git add" to track)

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Desktop/techie-horizon-05 (master)
$
```